
Memory-Efficient Pseudo-Labelling for Online Source-Free Universal Domain Adaptation using a Gaussian Mixture Model

Authors - Zilin Huang, Zihao Sheng, Yansong Qu, Junwei You, Sikai Chen
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Presented By:

Jay Fiske

Under the supervision of:

Udaya Kumar

Under the guidance of:

Prof. C. Krishna Mohan



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

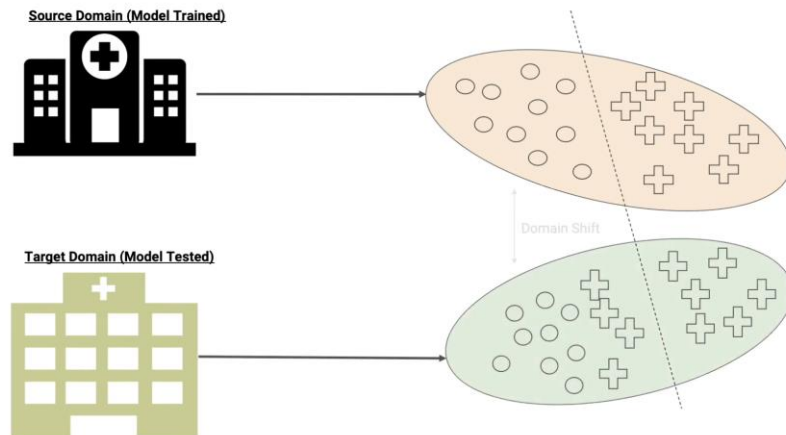
Introduction

In nature domain shift occurs:

- Distribution of feature changes
- Conditional relation between feature changes
- Sometimes, feature space changes as well

Some examples of domain shift:

- High resolution DSLR images vs low quality phone images
- Scanned images from different hospitals
- RGB images vs Infrared images

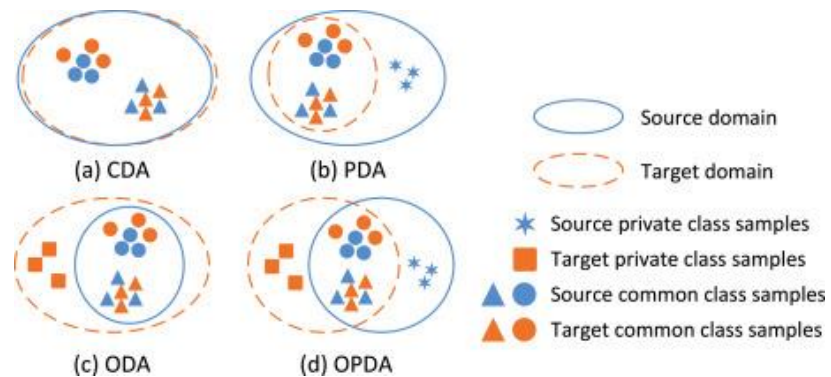


Introduction

- To tackle this problem domain adaptation was introduced, but recently new possibility was introduced namely, category shift
- In category shift, there is mismatch between source domain(dataset) and target domain(dataset)

Types of Category shifts:

- Closed domain Adaptation (CDA)
- Partial domain Adaptation (PDA)
- Open domain Adaptation (ODA)
- Open Partial Domain Adaptation (OPDA)



Motivation

- **Domain shifts are inevitable**

Real-world applications experience distribution shifts between training and deployment

- **Source Data is Often Inaccessible**

Due to privacy concerns, memory constraints, or proprietary data, source datasets cannot be accessed during adaptation. Only pre-trained models are available.

- **Category Shifts are Common**

Real scenarios involve new classes appearing and source classes disappearing.

- **Memory Efficiency is Critical**

For embedded systems and real-time applications, memory is limited. Existing methods require substantial memory (queues, teacher models).

Research Gap: No existing method simultaneously addresses all these constraints while maintaining high performance.

Problem Statement

Online Source-Free Universal Domain Adaptation (SF-UniDA):

Goal: Adapt a pre-trained source model to an unlabeled target data stream.

Three Key Paradigms:

1. **Universal:** Handle Partial-Set (PDA), Open-Set (ODA), and Open-Partial-Set (OPDA) scenarios without prior knowledge of the category shift.
2. **Source-Free:** No access to source data during adaptation; only the pre-trained model is available.
3. **Online:** Target data arrives in batches $\{x_{i,k}^t\}$ and is accessed only once.

New Constraint: Memory-Efficiency. Saving knowledge from previous batches must be done with minimal memory footprint

Challenges

- **Limited Data Representation:**
In online settings, individual batches cannot sufficiently represent the underlying data distribution, making pseudo-labeling difficult.
- **Catastrophic Forgetting:**
Without access to source data or past target data, models risk forgetting learned features.
- **Distinguishing Unknowns:**
The model must reliably reject samples from new classes (unknowns) while correctly classifying known classes.
- **Memory vs. Performance:** Storing previous samples (Memory Queue) or maintaining a model copy (Mean Teacher) requires substantial memory, which is scarce in embedded applications
- **Pseudo Labels:** Generating reliable pseudo labels online is challenging due to limited availability of data

Previous Approaches

1. Online Test-Time Adaptation (TTA):

- BatchNorm adaptation, entropy minimization, self-training
- Examples: TENT, SHOT, contrastive TTA
- Goal: Handle domain shift during deployment

Limitation:

- Assume closed-set (no new/missing classes)
- Fail under category shift (ODA / PDA / OPDA)

2. Universal Domain Adaptation (UDA)

- Clustering, prototype learning
- Handle both domain + category shift

Limitation:

- Require access to **source data**
- Not applicable in source-free or online settings

Previous Approaches (SF-UniDA methods)

3. Offline SF-UniDA Methods:

- **GLC / GLC++:** clustering + kNN + entropy rejection
- **LEAD:** feature decomposition into known / unknown
- Goal: Handle domain shift during deployment

Limitations:

- Depend on clustering or kNN
- Perform poorly in online settings
- Assume offline access to sufficient target data.

4. Online SF-UniDA Methods:

- **OWTTT:** dynamic prototypes + self-training
- **COMET:** contrastive mean-teacher framework

Limitation:

- Require **source prototypes** (not strictly source-free)
- Mean-teacher doubles model size → **high memory cost**
- Memory-heavy → impractical for embedded / real-time systems

Methodology

Core Objective: The goal is to achieve universal domain adaptation (DA) in an **online, source-free, and Memory efficient manner**.

Mathematical Framework:

- **Source Knowledge:** A neural network f is pre-trained on a source dataset $\mathcal{D}_s = \{\mathbf{x}_i^s \in \mathcal{X}_s, y_i^s \in \mathcal{Y}_s\}_{i=1}^{N_s}$
- **Target Data:** The model must adapt to unlabelled target data $\mathcal{D}_t = \{\mathbf{x}_i^t \in \mathcal{X}_t, ? \in \mathcal{Y}_t\}_{i=1}^{N_t}$
- **Model Architecture:** The network f is divided into a feature extractor g and a classifier h , expressed as $f = h \circ g$.

Methodology

1. GMM-Based Knowledge Transfer

Core Idea: Use a Gaussian Mixture Model (GMM) to capture the distribution of known classes in the feature space as a compact memory of past batches.

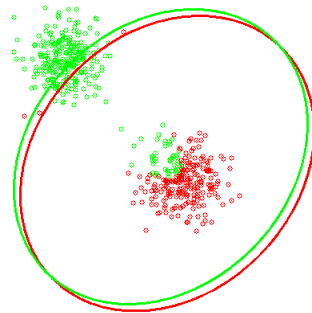
- **Iterative Updates:** Parameters (means μ_k and covariance Σ_k) are updated batch-wise via a weighted recalculation inspired by the EM algorithm.
- **Weighting Strategy:** Uses model predictions ($f(x)$) for intra-batch weighting and a class-wise sum of softmax outputs (s_k) for inter-batch weighting.
- **Decay Factor (α):** Controls the balance between retaining previous knowledge and adapting to new data.

$$s_k(c) = \alpha \cdot s_{k-1}(c) + \sum_{i=1}^{N_b} f_c(\mathbf{x}_{i,k}^t) ,$$

Methodology

$$\hat{\boldsymbol{\mu}}_k(c) = \frac{\alpha \cdot s_{k-1}(c) \cdot \hat{\boldsymbol{\mu}}_{k-1}(c) + \sum_{i=1}^{N_b} f_c(\mathbf{x}_{i,k}^t) \cdot r_{i,k}}{s_k(c)}$$

$$\hat{\boldsymbol{\Sigma}}_k(c) = \frac{\begin{pmatrix} \alpha \cdot s_{k-1}(c) \cdot \hat{\boldsymbol{\Sigma}}_{k-1}(c) + \sum_{i=1}^{N_b} f_c(\mathbf{x}_{i,k}^t) \\ \cdot (r_{i,k} - \hat{\boldsymbol{\mu}}_k(c)) (r_{i,k} - \hat{\boldsymbol{\mu}}_k(c))^T \end{pmatrix}}{s_k(c)}$$



Methodology

2. Pseudo-Labelling & OOD Detection

Strategy: Assign labels to known classes using the GMM while identifying "unknown" samples via Out-of-Distribution (OOD) detection.

- **Likelihood Estimation:** Calculates the probability of a sample belonging to each Gaussian distribution (known class).
- **Entropy-Based Rejection:** Uses the normalized Shannon entropy of GMM likelihoods to assess confidence.
- **Dual Thresholding:** Employs τ_k (known) and τ_u (unknown) to filter uncertain samples and prevent "negative transfer".

$$I(\mathbf{p}_{i,k}) = -\frac{1}{\log |\mathcal{Y}_s|} \cdot \mathbf{p}_{i,k}^T \cdot \log \mathbf{p}_{i,k}$$

$$\hat{y}_{i,k}^{\text{pl}} = \begin{cases} \arg \max_{c \in \mathcal{Y}_s} \mathbf{p}_{i,k} & I(\mathbf{p}_{i,k}) \leq \tau_k \\ |\mathcal{Y}_s| + 1 & I(\mathbf{p}_{i,k}) \geq \tau_u \\ \text{None} & \text{otherwise} \end{cases}$$

Methodology

3. Adaptation via Contrastive Loss

Assumption: The method is based on the assumption that the target data classes form unimodal gaussian distributed clusters in the feature space.

Contrastive loss is introduced which supports this hypothesis by enforcing the class-wise clustering of samples in the feature space.

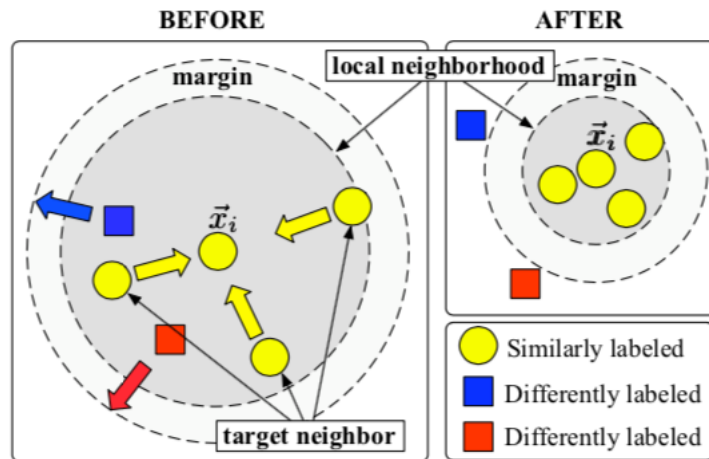
Function: Optimizes the feature extractor (g) to create a meaningful and well-separated feature space.

- **Positive Pairs:** Samples are pulled toward other samples of the same class and the corresponding GMM cluster center (mean).
- **Negative Pairs:** Distances are maximized between samples of different classes and between known samples and the "unknown" class.
- **Source-Free Advantage:** Uses GMM means as cluster centres instead of requiring source prototypes (like OWTTT and COMET-P)
- **Data Augmentation:** Each batch is doubled with augmented versions to ensure the feature space is robust to input variations.

Methodology

$$\mathcal{L}_C = - \sum_{j=1}^{2N_b} \sum_{i=1}^{2N_b} \mathbb{1}(\hat{y}_{j,k}^{\text{pl}} = \hat{y}_{i,k}^{\text{pl}}) \log \frac{\exp\left(\frac{\langle \tilde{r}_{j,k}, \tilde{r}_{i,k} \rangle}{\tau}\right)}{\sum_{l=1}^{2N_b} \exp\left(\frac{\langle \tilde{r}_{l,k}, \tilde{r}_{i,k} \rangle}{\tau}\right)}$$

$$- \sum_{c=1}^{|\mathcal{Y}_s|} \sum_{i=1}^{2N_b} \mathbb{1}(\hat{y}_{i,k}^{\text{pl}} = c) \log \frac{\exp\left(\frac{\langle \hat{\mu}_k(c), \tilde{r}_{i,k} \rangle}{\tau}\right)}{\sum_{l=1}^{2N_b} \exp\left(\frac{\langle \hat{\mu}_k(c), \tilde{r}_{l,k} \rangle}{\tau}\right)}$$



Methodology

4. Classifier Optimization via KL Divergence

Function: Trains the classifier (h) to handle feature representations that fall outside the original source training

- **Known Class Goal:** Maximizes the KL divergence between classifier output and a uniform distribution to encourage confident, "sharp" predictions.
- **Unknown Class Goal:** Minimizes the KL divergence for samples labelled as "unknown," forcing the output toward a uniform distribution (high entropy).
- **Joint Objective:** The final loss (L) combines the contrastive loss and KL divergence loss, weighted by a hyperparameter λ .

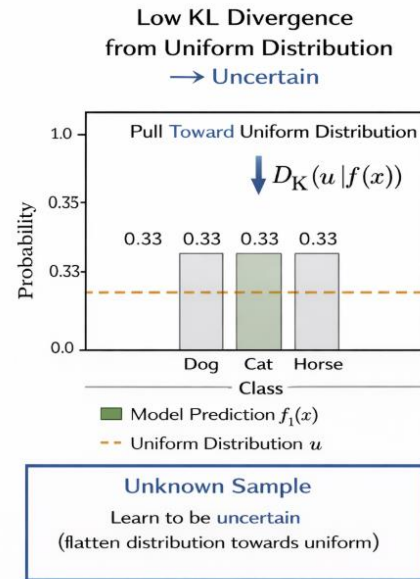
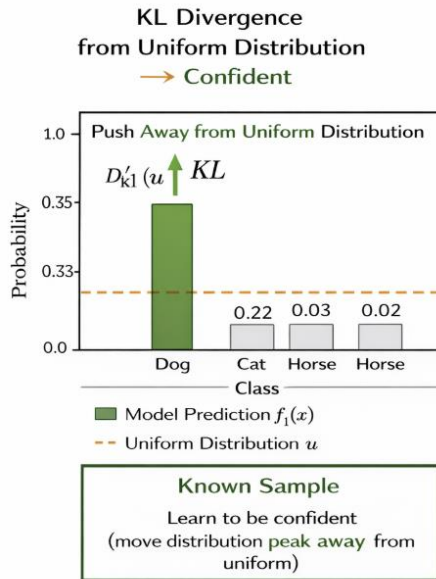
Methodology

$$\mathcal{L}_{\text{KLD}} = - \sum_{i=1}^{N_b} \mathbb{1} \left(\hat{y}_{i,k}^{\text{pl}} \in |\mathcal{Y}_s| \right) D_{\text{KL}} \left(\mathbf{u} || f(\mathbf{x}_{i,k}^t) \right) \\ + \sum_{i=1}^{N_b} \mathbb{1} \left(\hat{y}_{i,k}^{\text{pl}} \notin |\mathcal{Y}_s| \right) D_{\text{KL}} \left(\mathbf{u} || f(\mathbf{x}_{i,k}^t) \right)$$

where

$$D_{\text{KL}}(p||q) = - \sum_i p_i \log q_i + \sum_i p_i \log p_i$$

$$\mathcal{L} = \mathcal{L}_C + \lambda \mathcal{L}_{\text{KLD}}$$



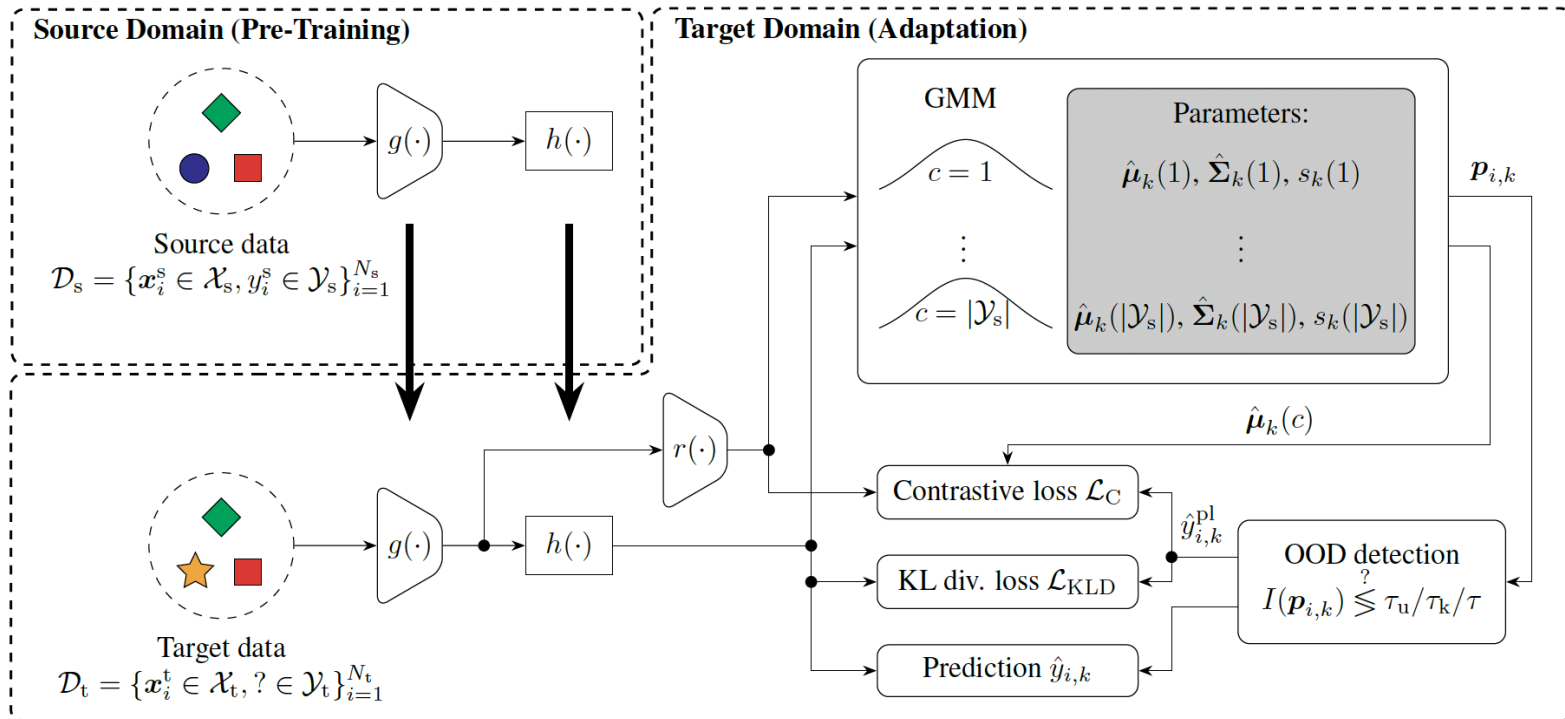
Methodology

5. Inference

- For inference, we combine standard prediction based on the maximum classifier output with the same OOD detection method used for pseudo-labeling.
- The only difference is that we now use a single threshold, τ , because every sample must receive a prediction.
- We set τ as the average of the two thresholds from pseudo-labeling: $\tau = \frac{\tau_k + \tau_u}{2}$
- The resulting prediction process for any sample $\mathbf{x}_{i,k}^t$ is as follows:

$$\hat{y}_{i,k} = \begin{cases} \arg \max f(\mathbf{x}_{i,k}^t) & I(\mathbf{p}_{i,k}) \leq \tau \\ |\mathcal{Y}_s| + 1 & I(\mathbf{p}_{i,k}) > \tau \end{cases}$$

Methodology



Memory Footprint & Efficiency Comparison

Objective: To evaluate the memory efficiency of the GMM-based knowledge transfer against existing state-of-the-art methods.

1. **GMM-based Method (Ours):** Stores a mean vector, symmetric covariance matrix, and weight for each of the

classes. Total parameters:
$$N_{\text{GMM}} = \left(FD_r + \frac{FD_r \cdot (FD_r + 1)}{2} + 1 \right) \cdot |\mathcal{Y}_s|$$

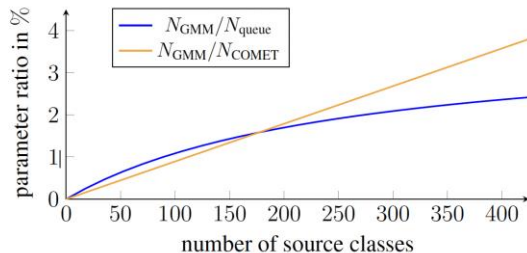
2. **Memory Queue:** Stores feature and prediction vectors for $L=55,388$ samples. Total parameters:

$$N_{\text{queue}} = L \cdot (FD + |\mathcal{Y}_s|)$$

3. **Mean Teacher (COMET):** Requires a full copy of the model architecture. For a ResNet-50 backbone, this is approximately $N_{\text{COMET}} \approx 24,000,000$ parameters.

Memory Footprint & Efficiency Comparison

- Even for $|Y_s| = 345$ source classes (PDA on DomainNet), our approach requires only approximately 2.2% of the memory needed by the memory queue and 3.1% compared to the mean teacher.
- This substantial reduction in memory usage makes the GMM-based approach **better suited for embedded applications** where memory resources are limited.



Results

(a) Accuracies in % for the PDA scenario

PDA	C2P	C2R	C2S	P2C	P2R	P2S	R2C	R2P	R2S	S2C	S2P	S2R	Avg.	V
Source-only	17.5	31.3	21.0	25.0	41.5	24.0	34.1	30.5	22.7	24.7	17.8	25.2	26.28	17.1
OWTTT [21]	22.6	28.9	25.1	26.3	28.3	24.6	27.2	26.6	22.9	27.6	19.9	14.5	24.54	28.1
COMET-P [39]	24.8	41.2	30.3	34.5	51.3	35.8	40.6	36.1	30.6	37.5	30.6	41.1	36.20	32.1
SHOT-P [22]	9.0	37.1	1.9	0.8	6.4	0.3	1.4	1.0	1.2	0.8	0.2	0.2	5.03	35.6
GLC [34]	12.9	13.4	16.9	18.1	12.1	16.7	16.8	11.0	13.4	23.4	16.5	12.3	15.29	8.2
GLC++ [35]	13.0	13.5	16.7	18.4	13.5	16.9	17.6	11.7	13.5	23.6	16.5	13.3	15.68	10.3
LEAD [33]	6.8	6.2	10.5	13.4	6.9	10.5	9.2	5.5	4.9	18.2	9.0	5.9	8.92	14.4
COMET-F [39]	23.4	38.8	28.3	32.2	48.9	33.7	40.4	36.5	30.7	34.8	28.0	38.3	34.50	31.7
GMM (Ours)	28.3	41.0	32.7	34.7	44.6	34.3	43.4	40.1	33.4	41.4	34.5	40.9	37.44	41.2

(b) H-scores in % for the ODA scenario

ODA	C2P	C2R	C2S	P2C	P2R	P2S	R2C	R2P	R2S	S2C	S2P	S2R	Avg.	V
Source-only	36.3	51.3	39.9	41.2	57.8	40.4	50.2	47.0	38.4	44.4	30.7	45.9	43.63	31.7
OWTTT [21]	42.1	41.9	45.0	45.3	47.0	42.9	48.7	35.9	40.8	51.9	45.4	44.1	44.25	56.7
COMET-P [39]	42.4	54.8	46.8	47.9	57.2	47.7	53.8	50.8	46.3	52.7	39.0	52.7	49.34	50.9
SHOT-O [22]	41.6	53.9	45.2	46.3	59.1	45.2	52.6	49.9	43.6	50.1	33.7	50.4	47.63	54.3
GLC [34]	30.6	34.4	35.8	32.4	26.2	30.3	30.1	23.8	22.5	43.1	32.1	30.1	30.95	8.6
GLC++ [35]	30.2	33.7	35.6	32.1	27.3	29.8	29.5	22.8	21.5	42.0	31.3	30.8	30.55	9.3
LEAD [33]	30.9	38.6	36.5	34.0	36.6	30.8	33.3	27.6	23.2	44.5	33.7	36.3	33.83	19.7
COMET-F [39]	40.7	53.2	45.0	46.2	57.2	46.2	53.0	50.3	45.3	50.0	37.0	51.3	47.95	49.0
GMM (Ours)	43.3	54.4	46.7	47.0	56.2	44.0	50.7	49.9	44.2	52.6	45.7	54.9	49.13	59.9

Results

(c) H-scores in % for the OPDA scenario

OPDA	C2P	C2R	C2S	P2C	P2R	P2S	R2C	R2P	R2S	S2C	S2P	S2R	Avg.	V
Source-only	39.5	55.4	42.7	42.0	57.6	38.8	51.7	47.6	38.6	46.8	32.2	47.9	45.07	26.9
OWTTT [21]	45.3	49.5	46.9	47.0	45.9	43.9	49.3	40.5	41.0	52.3	46.4	36.6	45.38	46.8
COMET-P [39]	45.6	58.6	48.9	47.8	57.7	46.8	55.5	51.5	46.2	54.1	39.0	54.2	50.49	42.9
SHOT-O [22]	44.8	57.9	47.8	47.2	59.3	43.7	54.0	50.1	42.8	52.0	35.3	52.3	48.93	45.7
GLC [34]	33.8	40.5	38.6	35.7	32.0	33.1	34.2	28.9	26.0	45.7	35.1	36.1	34.98	11.2
GLC++ [35]	33.4	39.9	38.3	35.0	33.5	32.6	33.6	28.1	25.7	45.1	34.7	35.7	34.63	14.1
LEAD [33]	33.3	42.0	37.4	35.0	37.4	31.1	33.5	30.1	25.2	45.6	34.9	37.9	35.28	23.8
COMET-F [39]	44.1	57.1	47.0	47.7	57.9	45.5	54.8	51.3	45.4	51.7	37.4	53.1	49.42	42.0
GMM (Ours)	46.4	57.6	48.4	48.5	57.1	44.2	51.9	50.9	44.8	53.9	46.7	56.3	50.56	60.3

Table 3. Accuracies (for PDA) and H-scores (for ODA and OPDA) in % for Office-Home. Best results are in red, second best in blue.

Source domain	PDA				ODA				OPDA			
	Ar	Cl	Pr	Re	Ar	Cl	Pr	Re	Ar	Cl	Pr	Re
Source-only	44.4	33.3	32.8	42.3	62.2	54.2	52.3	59.9	69.9	61.4	60.2	61.4
OWTTT [21]	34.0	28.0	29.8	32.2	58.9	61.0	57.2	57.4	69.2	71.8	64.2	65.3
COMET-P [39]	49.9	43.2	36.4	47.1	62.2	58.4	55.5	59.4	70.2	65.3	66.0	66.4
SHOT-O/P [22]	50.0	48.4	41.7	43.7	63.3	57.5	57.1	61.3	70.4	65.3	66.0	65.1
GLC [34]	21.1	20.3	18.0	20.2	37.0	40.4	37.8	38.2	51.1	52.6	50.5	52.3
GLC++ [35]	21.1	22.4	20.3	22.3	35.5	35.9	35.5	37.1	49.5	50.0	50.0	52.9
LEAD [33]	17.6	15.5	13.7	18.4	25.8	25.0	25.1	28.4	23.5	22.6	22.0	25.9
COMET-F [39]	49.1	40.7	36.1	45.4	61.8	57.5	55.1	58.9	69.7	64.3	64.7	65.0
GMM (Ours)	46.6	46.5	43.2	48.3	60.5	60.2	58.7	60.1	68.8	65.4	66.6	65.8

Conclusion

This work highlights **memory efficiency** as a critical yet previously overlooked challenge in **online source-free universal domain adaptation (SF-UniDA)**.

Proposed Solution:

- Introduced a **Gaussian Mixture Model (GMM)** to continuously capture the distribution of target data.
- Enables **memory-efficient knowledge transfer** across batches without storing past samples.
- Generates **reliable pseudo-labels** using maximum likelihood classification combined with **entropy-based OOD detection**.
- Achieves effective adaptation through a **contrastive loss + KL divergence loss**.

Conclusion

Results & Impact:

- Achieves **state-of-the-art or better performance** across experiments.
- Operates in a **strictly source-free** setting.
- Maintains **very low memory requirements**, making it suitable for **real-world and embedded applications**.

Final Takeaway:

- Modelling class distributions instead of storing data provides a scalable and practical path toward robust online domain adaptation

References

Pau Panareda Busto and Juergen Gall. Open set domain adaptation. In *2017 IEEE International Conference on Computer Vision (ICCV)*, pages 754–763, 2017. 2

Pau Panareda Busto, Ahsan Iqbal, and Juergen Gall. Open set domain adaptation for image and action recognition. *IEEE transactions on pattern analysis and machine intelligence*, 42(2):413–429, 2018. 2

Zhangjie Cao, Mingsheng Long, Jianmin Wang, and Michael I Jordan. Partial transfer learning with selective adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2724–2732, 2018. 2

Kaichao You, Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I Jordan. Universal domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2720–2729, 2019. 2

Jogendra Nath Kundu, Naveen Venkat, R Venkatesh Babu, et al. Universal source-free domain adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4544–4553, 2020. 2

Jogendra Nath Kundu, Naveen Venkat, Ambareesh Revanur, R Venkatesh Babu, et al. Towards inheritable models for open-set domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12376–12385, 2020. 2

Jogendra Nath Kundu, Naveen Venkat, Ambareesh Revanur, R Venkatesh Babu, et al. Towards inheritable models for open-set domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12376–12385, 2020. 2

Bo Fu, Zhangjie Cao, Mingsheng Long, and Jianmin Wang. Learning to detect open classes for universal domain adaptation. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XV 16*, pages 567–583. Springer, 2020. 2

Kuniaki Saito, Donghyun Kim, Stan Sclaroff, and Kate Saenko. Universal domain adaptation through self supervision. *Advances in neural information processing systems*, 33:16282–16292, 2020. 2

Kuniaki Saito and Kate Saenko. Ovanet: One-vs-all network for universal domain adaptation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9000–9009, 2021. 2

References

Dian Chen, Dequan Wang, Trevor Darrell, and Sayna Ebrahimi. Contrastive test-time adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 295–305, June 2022. 1, 2, 3, 6, 13

Mario Döbler, Robert A Marsden, and Bin Yang. Robust mean teacher for continual and gradual test-time adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7704–7714, 2023. 2

Pascal Schlachter and Bin Yang. Comet: Contrastive mean teacher for online source-free universal domain adaptation. In *International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2024. 1, 2, 3, 5, 6, 7, 8, 11, 13

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 8

Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. *Advances in neural information processing systems*, 33:18661–18673, 2020. 5

Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations*, 2021. 1, 2

Thank You !